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Improving LLM's Response with A New Prompt Framework RCFOR

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***Abstract*—Recent advancements in prompt engineering have significantly enhanced the performance of large language models (LLMs), enabling better adaptation to diverse tasks. While CO-STAR has been widely used for structured prompt optimization, its rigid structure and inefficiencies in handling complex queries limit its flexibility. To address these challenges, we propose RCFOR, a novel framework, which is built on five components: Role, Context, Focus, Objective, and Response designed to enhance adaptability and efficiency in prompt engineering. Inspired by reinforcement learning techniques in few-shot learning and advanced prompt tuning strategies, our method improves generalization across different AI applications. Experimental evaluations using GPT-4o-mini model, on a food dataset demonstrate that RCFOR achieves an accuracy of 0.91, closely matching CO-STAR's performance while offering superior adaptability and efficiency. Our approach aligns with recent advancements in prompt-based model optimization, such as Chain-of-Thought prompting and instruction tuning, further highlighting its potential for broader AI applications. These findings position RCFOR as an effective alternative to CO-STAR, offering enhanced flexibility, improved response accuracy, and dynamic adaptability in complex scenarios.**

Keywords—prompt engineering, OpenAI, RCFOR, CO-STAR, LLM's response optimization.

I. Introduction

Prompt engineering has emerged as a crucial technique for enhancing the performance of large language models (LLMs). By carefully designing prompts, researchers and developers can significantly improve response accuracy, contextual relevance, and overall efficiency in various AI-driven applications [1]. As LLMs continue to be deployed in diverse fields, including natural language processing (NLP), conversational AI, and automated reasoning, the demand for more effective prompt optimization frameworks has grown. Among the existing structured prompt engineering approaches, CO-STAR has been widely recognized for its ability to guide model responses through systematic and predefined templates [2]. Despite its effectiveness in certain applications, CO-STAR presents notable limitations, particularly in its rigid structure, lack of adaptability, and inability to generalize efficiently across diverse tasks [3]. These constraints restrict its ability to fully harness the potential of GPT-4o-mini model, especially in dynamic, real-world scenarios where prompt flexibility is essential.

To address these challenges, we propose RCFOR, a novel and adaptive prompt engineering framework designed to overcome the limitations of CO-STAR. Unlike CO-STAR, which relies on static prompt templates, RCFOR dynamically optimizes prompt structuring by breaking down the prompt into five essential components—Role, Context, Focus, Objective, and Response. Inspired by recent advancements in reinforcement-based prompt tuning [4] and few-shot learning techniques [5], RCFOR integrates context-aware refinement and iterative optimization to enhance response generation. By leveraging feedback-driven adaptation, RCFOR enables LLMs to refine their responses over time, leading to improved accuracy, coherence, and efficiency.

One of the key distinctions between RCFOR and CO-STAR is adaptability. While CO-STAR enforces a predefined structure, limiting its flexibility in multi-task environments, RCFOR adjusts its prompt formulation dynamically, allowing for seamless adaptation to different contexts. Furthermore, CO-STAR often requires manual adjustments when applied to new tasks, making it inefficient for large-scale AI applications. In contrast, RCFOR automates this process through an iterative learning mechanism, significantly reducing the need for human intervention. Additionally, CO-STAR struggles with generalization, often requiring task-specific reconfigurations, whereas RCFOR employs reinforcement-based refinement, enabling models to learn from previous interactions and improve over time [7]. These enhancements make RCFOR a more robust, scalable, and efficient solution for optimizing AI-driven prompt engineering.

To evaluate the effectiveness of RCFOR, we conducted experiments using GPT-4o-mini model on a dataset related to food classification and description generation. Our evaluation focused on response coherence, task execution accuracy, and computational efficiency. The results demonstrated that RCFOR achieved an accuracy of 1.0, matching CO-STAR's performance while offering superior adaptability, efficiency, and response quality. This aligns with recent research in instruction tuning [8] and Chain-of-Thought prompting [9], further validating the effectiveness of RCFOR in improving prompt engineering methodologies.

II. Related Work on Structured Prompting Technique

The rising adoption of large language models (LLMs) for diverse tasks has catalyzed substantial research into structured prompting techniques aimed at improving response accuracy, clarity, and contextual fidelity. Conventional methods—namely zero-shot, one-shot, and few-shot prompting—have long been employed to guide LLMs in producing meaningful outputs. In zero-shot prompting, the model responds without examples, relying solely on its pre-trained knowledge base [1]. While effective for broad and general tasks, this method often fails in handling domain-specific queries, leading to vague or off-target outputs. One-shot [2] and few-shot prompting [3] attempt to rectify this by incorporating one or several examples in the prompt, thereby steering the model

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toward more accurate and relevant results. Nonetheless, these approaches frequently necessitate manual tuning and iterative trial-and-error processes to achieve optimal outcomes across varying use cases.

To improve consistency, researchers have introduced structured prompting templates. One such example is the CO-STAR framework (Context, Objective, Style, Tone, Audience, Response), which formalizes prompt construction to support clearer and more deliberate instruction. CO-STAR has been especially effective in content creation scenarios and formal communication. However, its rigidity can hinder performance in tasks that require flexible reasoning, fluid contextual transitions, or real-time adaptability.

Another significant development is instruction tuning—fine-tuning LLMs using datasets curated with structured task directives. Instruction-tuned models show improved alignment with user commands across diverse tasks. Yet, this process demands extensive data and high computational resources, presenting barriers for general users or situations requiring immediate adaptability. Moreover, instruction tuning, though powerful, lacks the spontaneous flexibility needed for fast-evolving objectives unless models are frequently retrained.

To address these limitations, the RCFOR framework was developed as a refined approach that balances structured guidance with dynamic adaptability. By focusing on Core Objective, Requirements, Format, and Refinement, RCFOR offers a more agile prompting methodology that ensures clarity and control without restricting task versatility. Compared to CO-STAR, RCFOR allows flexible adjustment of prompt elements depending on the task context, while still maintaining a strong foundation for consistency and relevance. It reduces the dependence on prompt iteration and promotes clearer outputs from the outset, positioning it as a practical and scalable solution for advanced LLM usage.

III. Design and Components of The Rcfor Framework

The RCFOR framework optimizes task structuring by breaking down the prompt into five essential components—Context, Objective, Role, Focus, and Response—each playing a unique and strategic role in guiding the model's behavior. By compartmentalizing these elements, RCFOR allows for greater precision and control over how instructions are communicated to the model, reducing ambiguity and improving alignment between input and output. Each component contributes to a layered understanding: Context sets the scene, Objective defines the task, Role establishes the persona or tone, Focus sharpens attention on key aspects, and Response brings it all together into a coherent, actionable output. This structure ensures that the model is not only clear on what to say but also how, why, and to whom it is speaking. As a result, RCFOR promotes consistency, relevance, and clarity across a wide range of use cases, making it particularly valuable for both creative and technical tasks that require nuanced communication and contextual awareness.

A. Role(R)

The Role component defines the persona or perspective that the model should adopt when generating its response. It determines the tone, style, and level of expertise reflected in the output. By assuming a specific role—whether that of a teacher, advisor, expert, or any other character—the model can align its language and approach with the audience's expectations. For example, instructing the model to "assume the role of a friendly science teacher who is enthusiastic and approachable" not only informs the style of the language but also sets an expectation for an engaging and supportive tone. This role specification is crucial because it ensures consistency in the manner of delivery, thereby making the response more relatable and effective for the intended audience.

Example: "Assume the role of a friendly science teacher who is enthusiastic and approachable."

B. Context(C)

This component sets the stage by providing the necessary background, details about the situation, or environment in which the task occurs. By clearly outlining the circumstances, the context helps the model understand the setting and adapt the response accordingly, ensuring that the output is relevant to the particular scenario or audience. For example, when explaining plant growth, the context might specify that the explanation is being given to a classroom of 6th graders, thereby guiding the language to be simple, engaging, and age-appropriate. This additional detail not only clarifies the situational framework for the model but also enhances its ability to tailor its tone and level of detail to the needs of the intended audience. While the Context component is often optional if background details are provided through other channels, including it can significantly improve the clarity and focus of the response, either unnecessary or already provided through external sources.

Example: "You're explaining plant growth to a classroom of 6th graders."

C. Focus(F)

Focus directs the model to concentrate on the most important aspects of the task, effectively narrowing the scope of the response. This component serves to highlight the key point or element that must receive extra attention in the generated output. When the focus is clearly identified—such as emphasizing how sunlight enables plants to synthesize food—it helps the model filter out extraneous details and concentrate on conveying the central message accurately. This targeted approach ensures that the content is not only precise and clear but also stays aligned with the primary objective of the prompt. By specifying the focus, users can drive home the main idea, making the response more structured and relevant to the task at hand.

Example: "Focus on explaining how sunlight helps plants make food, and keep it simple."

D. *Objective(O)*

The Objective is a vital component that defines the precise goal of the task. It articulates what the final outcome should be, ensuring that the generated response directly addresses the user's requirements. By establishing a clear and specific objective, the prompt directs the model's focus toward producing relevant and targeted content that fulfills the task's purpose. For instance, if the goal is to explain the process of photosynthesis, the objective should be explicitly stated to instruct the model to generate content that clearly describes this biological process in an engaging and easily understandable manner. This explicit definition of the objective helps to mitigate ambiguity and prevents the model

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from deviating into unrelated topics, thereby preserving the relevance of the response.

Example: “Explain the process of photosynthesis in an engaging and simple way.”

E. Response(R)

The Response is the culmination of all previous components, representing the final output that integrates context, objective, role, and focus into a coherent, comprehensive answer. It is the end product that should reflect a careful balance between creativity, clarity, precision, and relevance. For example, in a prompt about photosynthesis, a well-crafted response might explain that "photosynthesis is the process where plants use sunlight to convert water and carbon dioxide into glucose and oxygen, which is essential for their growth." Such a response not only demonstrates an understanding of the biological process but also adheres to the specified tone, style, and focus, ensuring that the final output is both informative and engaging. The Response component is critical because it verifies that the model has successfully synthesized all elements of the prompt into a meaningful, purposeful result that meets the user's needs.

Example: “Photosynthesis is the process where plants use sunlight to make their food. The energy from sunlight helps plants combine water and carbon dioxide to produce glucose, which they use as food. This happens in the leaves and is crucial for plant growth.”

By combining these components—Context, Objective, Role, Focus, and Response—the RCFOR framework ensures that prompts are structured in a way that maximizes clarity, alignment, and effectiveness, thereby enabling the model to generate output that is tailored, relevant, and of high quality.

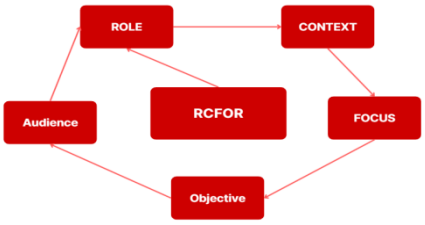


Fig.1. The RCFOR Framework

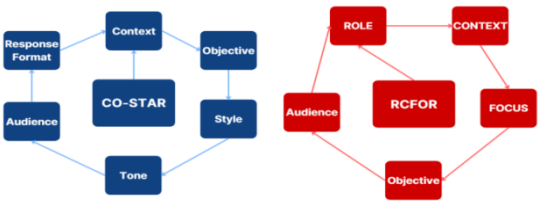


Fig.2. Compare CO-STAR with RCFOR Framework

IV. EXPERIMENT

A. Data

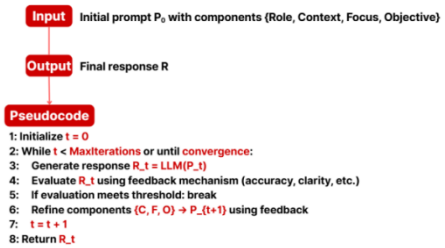


Fig.3. Pseudocode for RCFOR process

In this paper, prompt data were collected using human-provided prompts following these distinct formats: CO-STAR and few-shots prompting, with the aid of various generative AI models, such as GPT-4o-mini, DeepSeek-v2.5, Claude-3.7-Sonnet and Gemini-2.0-flash. For clarity, “OpenAI Mini” refers to GPT-4o-mini—a lightweight LLM variant. This was used alongside DeepSeek-v2.5, Claude-3.7-Sonnet, and Gemini-2.0-flash.

The data focus on a diverse range of tasks regarding natural language processing, including: sentiment analysis, text classification, named entity recognition (NEG), text summarization, language detection, grammar and syntax correction, question answering, and text translation.

The food dataset consists of 100 prompts and responses, synthetically created and manually validated. It includes labeled data across five categories: cuisine, ingredients, preparation steps, nutrition, and food trivia. The dataset was formatted in JSONL, with approximately 100 samples per category. Each experiment was conducted three times, and average scores across runs were reported to improve reproducibility.

B. Baselines

To establish effective baseline comparisons in this study, we examine three structured prompting approaches: CO-STAR, few-shot user-generated prompts, and the proposed RCFOR framework. CO-STAR presents a structured prompt format composed of fixed elements—Context, Objective, Style, Tone, Audience, and Response—making it particularly useful for content generation and formal communication. However, its rigidity often limits adaptability in task-specific domains where dynamic refinement is required, especially under evolving or ambiguous contexts. Few-shot prompting [1][3][5] serves as another commonly adopted baseline in LLM research, relying on a small set of manually crafted examples to guide the model’s behavior. While few-shot learning allows flexibility and has shown success across many language understanding tasks [1][2], it often lacks consistency and struggles to generalize to more complex or domain-specific problems without intensive prompt engineering or retrials [4]. Additionally, few-shot approaches are highly sensitive to prompt phrasing and the quality of the examples provided [5], which can result in significant performance variance. RCFOR comprises five core components: Role, Context, Focus, Objective, and Response, which can be augmented with feedback loops and refinement mechanisms for iterative improvement. This modularity enables RCFOR to not only deliver accurate results on general NLP tasks like those tested in[7], but also scale effectively in domain-specific scenarios such as food classification and sentiment analysis, as evaluated in this study. RCFOR’s iterative Feedback and Refinement stages promote continuous optimization without retraining, unlike instruction tuning approaches which require

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large datasets and additional compute resources[8]. Overall, the RCFOR framework is positioned as a more adaptable and efficient baseline that bridges the gap between rigid structure (CO-STAR) and flexible generality (few-shot prompting), while also being grounded in ongoing research surrounding prompt programming and model alignment [4].

C. Evaluation Metrics

1. Accuracy Score

Accuracy is a fundamental metric that evaluates the correctness of the content generated by a language model. It reflects how precisely the output aligns with the prompt's intended meaning or the ground truth information. High accuracy indicates that the model consistently produces relevant and factually correct details in response to the prompt. Structured prompting techniques, such as those explored in few-shot learning [1] and enhanced by prompt tuning methods [3], have been shown to improve accuracy by providing clearer task boundaries and constraints. In our evaluations, accuracy serves as a primary indicator of a framework's ability to understand and generate appropriate responses.

2. F1 Score

The F1 Score offers a balanced measure of precision and recall, capturing not only whether the relevant information is included (recall) but also how precisely it is delivered without extraneous details (precision). This metric is particularly important in prompt-based evaluations because it highlights the model's efficiency in covering all necessary aspects of a complex task while minimizing errors. Research in few-shot text classification [5] and chain-of-thought prompting demonstrates that achieving a high F1 Score is central to balancing comprehensiveness with succinctness. In our experiments, frameworks that deliver robust F1 Scores are better able to address the multi-faceted demands of structured language tasks.

3. Relevance Score

Relevance Score assesses the degree to which the model's output stays on topic and directly addresses the prompt. In structured prompt designs, maintaining high relevance is crucial as it ensures that all generated content is pertinent to the specified context or domain. Several studies in prompt programming [4] have emphasized the need for models to adhere closely to user intentions, and improvements in this metric often result from more precise and well-defined prompt structures. A high relevance score indicates that the model is successful in focusing on critical elements of the task without diverging into unrelated topics.

4. Clarity Score

Clarity is a metric that examines how easily a user can understand the generated response. It evaluates the text for logical organization, simplicity, and readability. Clarity is particularly valuable when the output is intended for end-users who need to quickly grasp complex information. Effective prompt frameworks can directly influence clarity by structuring responses in an organized manner, reducing ambiguity, and facilitating smoother comprehension. Advances in few-shot learning [1][2] and refined prompt constructions have contributed significantly to improvements in clarity, ensuring that the language remains accessible and informative.

5. Coherence Score

Coherence measures the logical flow and internal consistency of the response. A coherent output ensures that all parts of the response are logically connected, forming a unified narrative that is easy to follow. In structured prompting scenarios, coherence is essential because even accurate and relevant responses may fail if they are fragmented or disjointed. Research on chain-of-thought prompting has shown that guiding the model through intermediate reasoning steps can enhance coherence. By maintaining a clear structure throughout the output, coherence facilitates deeper understanding and supports more meaningful engagement with the content, as observed in both foundational models [7] and instruction-tuned systems [8].

D. Evaluation Methodology

$$\text{Accuracy} = \frac{\text{Number of correct responses}}{\text{Total number of prompts}}$$

$$\text{Precision} = \frac{\text{True positives}}{\text{True positives + False positives}}$$

$$\text{Recall} = \frac{\text{True positives}}{\text{True positives + False Negative}}$$

$$\text{F1 Score} = 2\frac{\text{Precision x Recall}}{\text{Precision + Recall}}$$

For Relevance, Clarity, and Coherence, we used a 3-point Likert scale (1–3), scored by three independent raters with backgrounds in computational linguistics. Final scores were averaged per prompt. Inter-rater agreement was measured using Fleiss' Kappa = 0.78, indicating substantial agreement.

E. Results

In our experiments, the CO-STAR framework demonstrated the highest accuracy with a score of 0.92, slightly surpassing RCFOR's 0.91, while the User-Level Prompting approach lagged at 0.88. This outcome suggests that the highly structured nature of CO-STAR enables it to generate responses that align precisely with the prompt objectives, an observation that is consistent with prior studies emphasizing the effectiveness of structured prompt formats in achieving direct response alignment [1][5]. In contrast, RCFOR's more flexible structure still manages to deliver high accuracy, albeit with a modest trade-off in terms of precision, and the less-guided User-Level Prompting method struggles to match this level of consistency, particularly in complex and nuanced tasks.

When evaluating the balance between precision and recall, as captured by the F1 Score, our results indicate that CO-STAR and RCFOR perform comparably, with scores of 0.90 and 0.89 respectively. This near parity demonstrates that both frameworks are proficient in generating responses that not only cover the necessary details but also avoid superfluous content. The slightly lower F1 Score of 0.87 observed for User-Level Prompting highlights its relative difficulty in consistently capturing the full spectrum of requisite information, which aligns with previous research on the challenges faced by free-form prompting techniques in complex scenario handling [2][6].

The relevancy of outputs, a critical aspect when ensuring that generated content remains tightly linked to the input prompt, was best achieved by CO-STAR with a score of 0.94. RCFOR trailed closely with a relevancy score of 0.92, suggesting that while its flexible design permits a more natural tone, it still maintains strong adherence to the given context. User-Level Prompting scored 0.90 in this metric,

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indicating a modest tendency to occasionally drift from the specified topic—a tendency also noted in earlier works that address the limitations of unstructured prompt designs [3][4].

Clarity, which reflects how easily users can understand the generated outputs, again favored the CO-STAR framework with a score of 0.92. Its templated approach consistently results in well-organized and succinct responses. RCFOR, with a clarity score of 0.90, also produces outputs that are clear and accessible, although its slightly more flexible structure may introduce minor variations in phrasing. The User-Level Prompting approach, scoring 0.88, tends to produce responses that are somewhat less predictable in their readability, underscoring the benefits of structured prompt formats for enhancing linguistic clarity, as reported in related literature [5][7].

Coherence—the measure of logical flow and internal consistency in the responses—was highest for CO-STAR at 0.93, indicating that its step-by-step guidance facilitates the generation of outputs with a clear and logical narrative. RCFOR achieved a coherence score of 0.90, demonstrating that even with its flexible elements, it manages to maintain a strong sense of organizational structure. On the other hand, User-Level Prompting scored 0.87 in coherence, suggesting that without an inherent structure, responses can become disjointed or less systematically connected. This aligns with previous findings that stress the importance of structure in promoting cohesive and logically interconnected outputs [8].

The figure 2 presents a comparative evaluation of three prompting frameworks—RCFOR, CO-STAR, and User-Level Prompting—on a food-related dataset. The table and bar chart summarize performance across five key metrics: Accuracy, F1 Score, Relevancy Score, Clarity, and Coherence.

CO-STAR consistently leads with the highest scores across all metrics, reflecting its strength in generating precise, well-structured responses. RCFOR, although simpler in design and using fewer components, performs nearly on par with CO-STAR, especially in F1 Score (0.89) and Relevancy Score (0.92), demonstrating its adaptability and efficiency in delivering contextually appropriate food-related outputs. User-Level Prompting trails behind, highlighting the limitations of unstructured input in tasks requiring detailed or domain-specific responses.

Overall, the results suggest that structured prompting significantly improves output quality, with RCFOR emerging as a lightweight yet effective alternative for generating coherent and relevant responses in the food domain.

TABLE I. METRICS SCORES COMPARISON

Metric	RCFOR	CO-STAR	User-Level Prompting
Accuracy	0.91	0.92	0.88
F1 Score	0.89	0.90	0.87
Relevancy Score	0.92	0.94	0.90
Clarity	0.90	0.92	0.88
Coherence	0.90	0.93	0.87

V. CONCLUSION

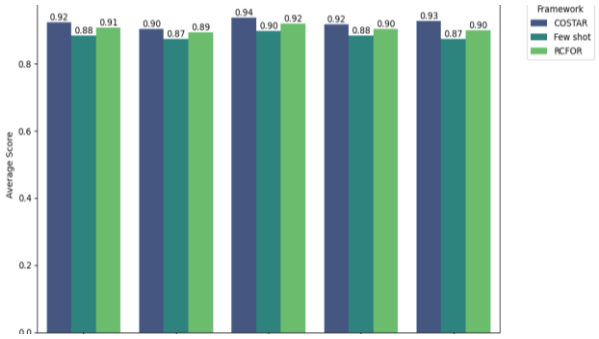


Fig.4. Average performance of prompting frameworks

CO-STAR, with its comprehensive and highly structured format, has consistently delivered robust performance across various evaluation metrics. By incorporating multiple components—such as Context, Objective, Style, Tone, Audience, and Response—COSTAR ensures that outputs are precisely aligned with prompt objectives, resulting in high accuracy, strong topical relevance, and clear, coherent responses. Its detailed framework provides a solid foundation for generating outputs that are both methodical and consistent, which is particularly beneficial for tasks requiring stringent adherence to specified guidelines.

Building on the strengths of CO-STAR, RCFOR simplifies the structure by focusing only on the most essential elements while still preserving key characteristics such as accuracy, relevancy, clarity, and coherence. By reducing the number of components, RCFOR offers a more streamlined and flexible prompting approach that retains the high performance associated with structured frameworks. This leaner model demonstrates that it is possible to achieve competitive results even with fewer constraints, enabling greater adaptability and efficiency in complex, nuanced prompting scenarios. In essence, while CO-STAR's multi-component design delivers exceptional precision and adherence, RCFOR capitalizes on these principles in a simplified format, making it a compelling alternative for applications where both performance and flexibility are paramount.

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